

POWER ELECTRONICS

Power Parameter
Calculations

Outcomes

- Determine the equation for various types of waveforms
- Calculate average value of each type
- Calculate the root-mean-squared value of each type
- Define instantaneous and average power
- Calculate the average power delivered by variety of waveforms.

EQUATIONS

Of Common Waveforms Involved in Power Electronics

DC Waveform

- Simplest waveform
- Voltage from the power supplies used to bias the electronic circuits
- Constant value
- Remains equal to the peak value



$$v(t) = V_p$$

Sinusoidal Waveform

- Most common test signal
- Circuit's effect on phase and amplitude (no effect on shape by linear components)
- Utility power

$$v(t) = V_P \sin \omega t$$

- Or

$$v(\theta) = V_p \sin \theta$$

- Equation for inverted sine wave

$$v(\theta) = -V_p \sin \theta$$

Or

$$v(\theta) = -V_p \sin(\theta + \pi)$$

Or

$$v(\theta) = V_p \sin(\theta - \pi)$$

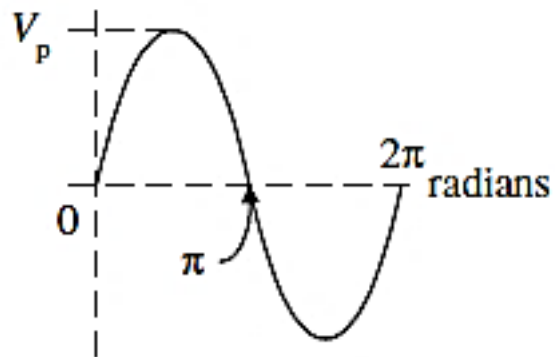
Effect of Capacitor and Inductor

- Purely capacitive circuit causes the current to lead the voltage by 90° ($\pi/2$) radians

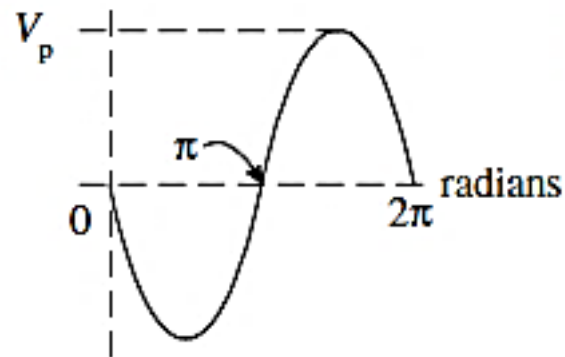
$$i(\theta) = I_P \sin\left(\theta + \frac{\pi}{2}\right)$$

- Purely inductive circuit causes the current to lag the voltage by -90° ($-\pi/2$) radians

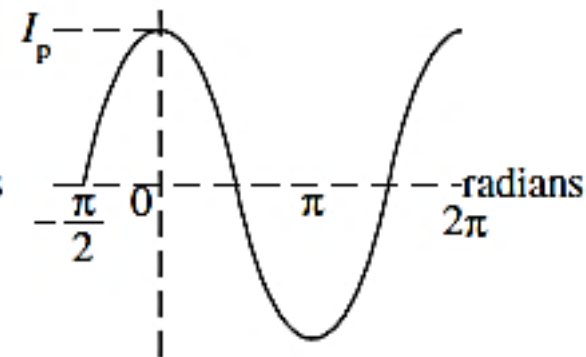
$$i(\theta) = I_P \sin\left(\theta - \frac{\pi}{2}\right)$$



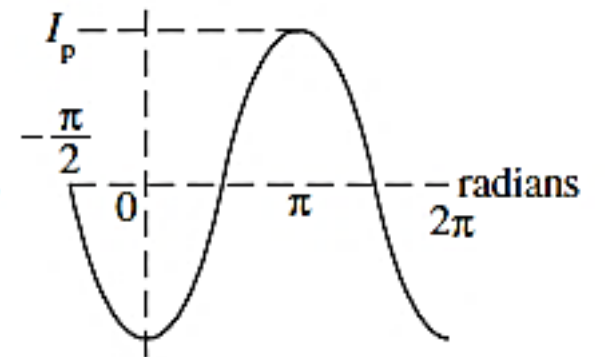
(a) Simple sine wave



(b) Inverted sine wave



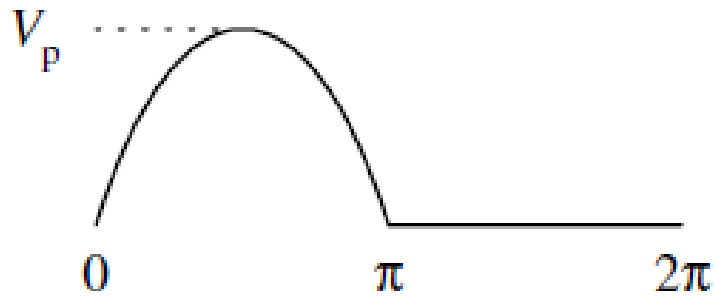
(c) Sine wave leading by 90°



(d) Sine wave lagging by 90°

Half-Wave Rectified Sine Wave

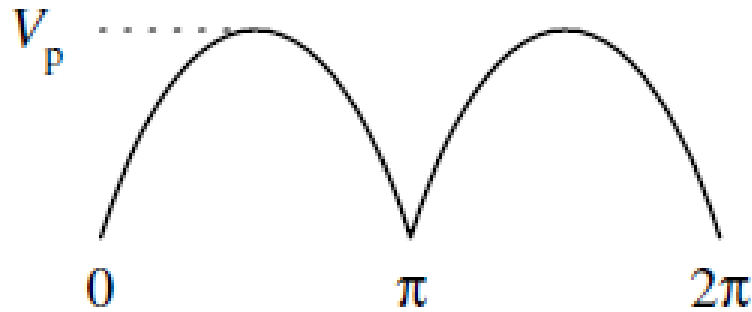
- AC to DC conversion
- Simplest form: single diode



$$v(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$$

Full-Wave Rectified Sine Wave

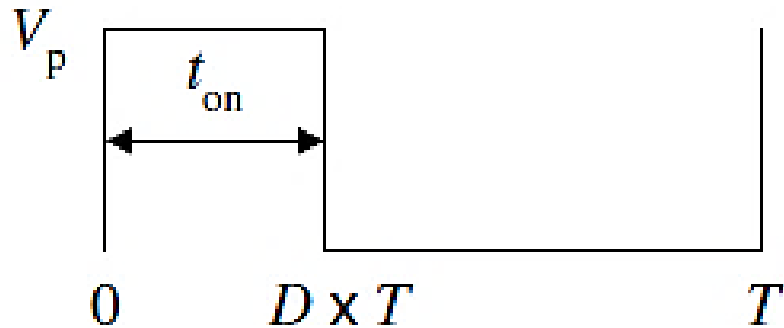
- AC to DC conversion
- Simplest form: two diodes with center tapped transformer
- Second part of waveform is inverted



$$v(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ -V_p \sin \theta & \pi \leq \theta < 2\pi \end{cases}$$

Rectangular Wave

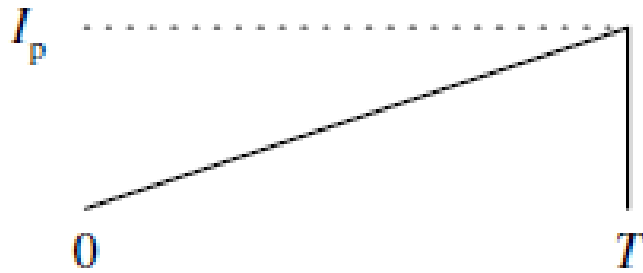
- Switching power supplies
- Allow user to step voltage up or down
- Typically 90% efficient (negligible energy loss, negligible heat generation)
- Same concept is extended to dc motor drives
- Duty cycle $D = \frac{t_{on}}{T}$



$$v(t) = \begin{cases} V_p & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

Triangular Wave

- Inductive load (motor) or inductor within a filter
- Opposes rapid current changes
- Simplest triangle is $y = mx + b$
- Equation for a ramp signal



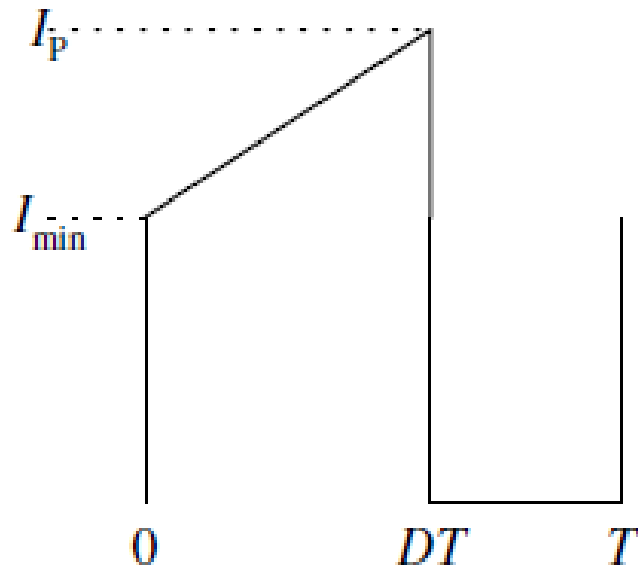
$$i(t) = \frac{I_P}{T} t$$

Example 1:

- Write the equation for the ramp that has a peak current of 3 A and a frequency of 150 kHz.
- Calculate the current 5 μs after the beginning of the signal.

Trapezoid Waveform

- More practical but a bit complicated



$$i(t) = \begin{cases} \frac{I_p - I_{\min}}{DT} t + I_{\min} & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

Example 2

- Write equation for a trapezoid with a frequency of 500 kHz, a 40% duty cycle, a peak current of 2 A, and a ripple of 300 mA.
- Find the current at 400 ns and at 1 μ s.

AVERAGE VALUE

Calculations

Average Value of DC Waveform

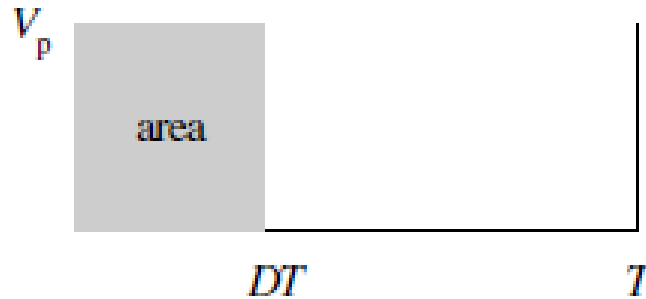
- Also known as dc value
- Equal to peak value of the dc signal

$$V_{dc} = V_{avg} = V_p$$

Average Value of Rectangular Waveform

- If signal is 10 V for half of the time, average is 5 V
- If signal is 10 V for 20%, the average is 2 V

$$V_{avg} = DV_P$$



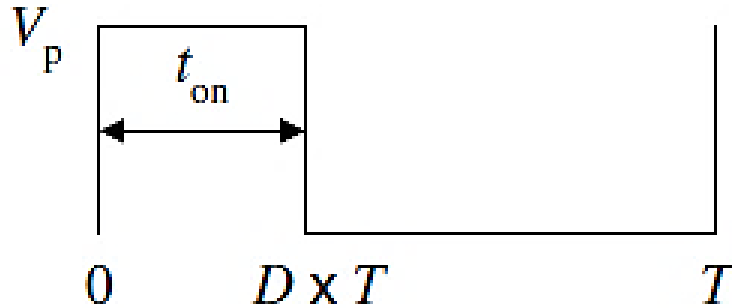
General Equation for Average Value

- Integrator equation

$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt$$

Average Value of Rectangular Waveform

- For rectangular wave:



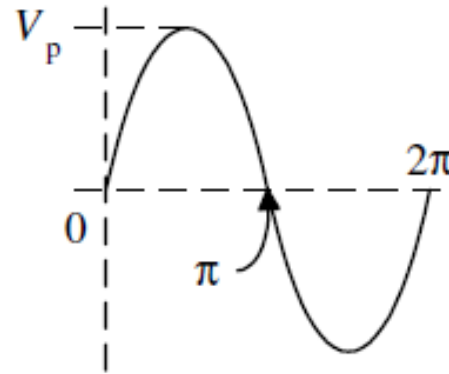
$$v(t) = \begin{cases} V_p & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$V_{ave} = \frac{1}{T} \left(\int_0^{DT} V_p dt + \int_{DT}^T 0 dt \right)$$

Average Value of Sinusoids

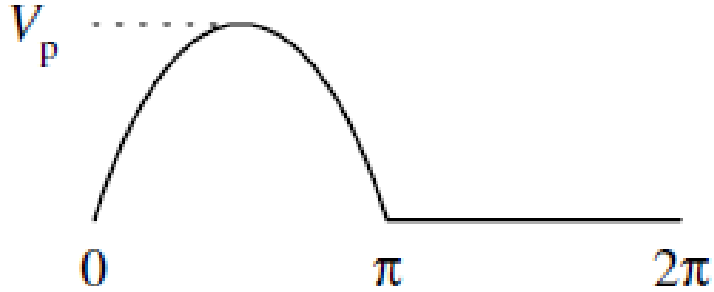
- Average value of a sinusoid is 0

$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt$$



Average Value of Half-Wave Rectified Sine

- Non zero average value

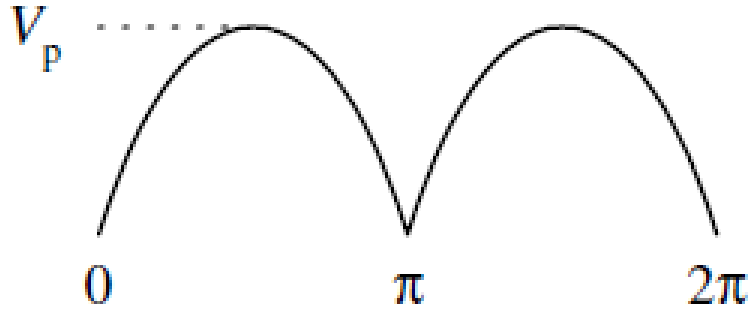


$$v(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$$

$$V_{\text{ave}} = \frac{1}{2\pi} \left(\int_0^{\pi} V_p \sin \theta \, d\theta + \int_{\pi}^{2\pi} 0 \, d\theta \right)$$

Average Value of Full-Wave Rectified Sine

- Non zero average value

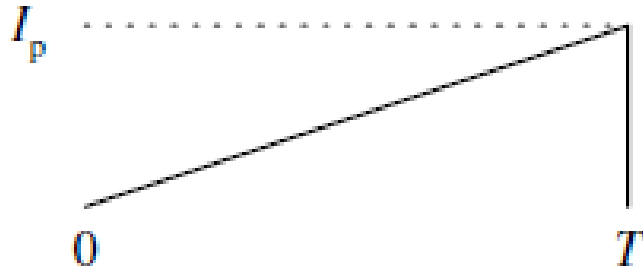


$$v(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ -V_p \sin \theta & \pi \leq \theta < 2\pi \end{cases}$$

$$V_{\text{ave}} = \frac{1}{2\pi} \left(\int_0^{\pi} V_p \sin \theta \, d\theta + \int_{\pi}^{2\pi} -V_p \sin \theta \, d\theta \right)$$

Average Value of Triangular Wave

- Half of the peak value

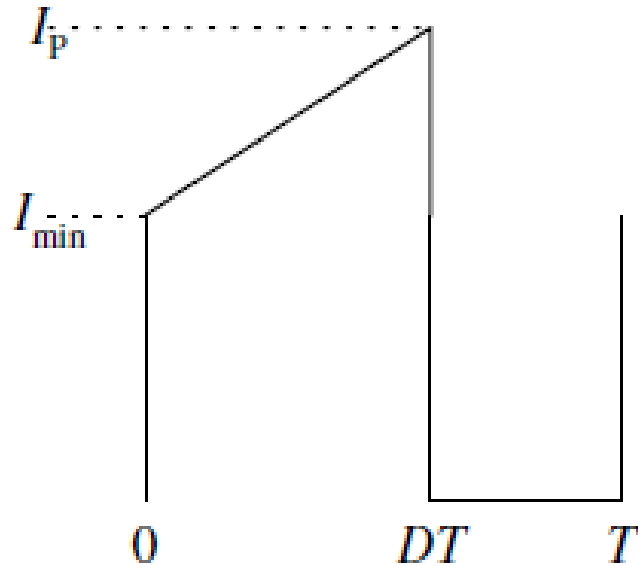


$$i(t) = \frac{I_p}{T} t$$

$$I_{ave} = \frac{1}{T} \int_0^T \frac{I_p}{T} t dt$$

Average Value of Trapezoidal Wave

- Similar approach



$$i(t) = \begin{cases} \frac{I_p - I_{\min}}{DT} t + I_{\min} & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$I_{\text{ave}} = \frac{1}{T} \left[\int_0^{DT} \left(\frac{I_p - I_{\min}}{DT} t + I_{\min} \right) dt \right] + \int_{DT}^T 0 dt$$

Example 3

Calculate the average value of a trapezoid with the following parameters:

Frequency: 100 kHz

Duty cycle: 35%

$$I_P = 3 \text{ A}$$

$$I_{ripple} = 400 \text{ mA}$$

Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan